ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

Diploma (CSE)

Semester: IV

SUBJECT CODE: DCS402

OOP'S WITH C++

Time: 03:00 Hrs. **Maximum Marks: 60**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following questions. **(12 Marks)**

Q. 1. Answer the following questions: **(1×6 = 6 Marks)**

(i) **Define** Object-Oriented Programming.

(ii) **List** the four basic principles of OOP.

(iii) **State** the purpose of a constructor in C++.

(iv) **Mention** any two types of inheritance in C++.

(v) **Name** any two access specifiers used in C++.

(vi) **Draw** a simple class diagram showing inheritance.

Q. 2. Answer the following questions: **(1×6 = 6 Marks)**

(i) **Paraphrase** the concept of polymorphism in your own words.

(ii) **Clarify** the difference between function overloading and function overriding.

(iii) **Translate** the idea of encapsulation into a real-world example.

(iv) **Represent** the relationship between a base class and derived class with a diagram.

(v) **Clarify** the use of constructors and destructors in managing object lifecycle.

(vi) **Paraphrase** the role of operator overloading in C++.

SECTION – B

NOTE:Attempt all questions. (6×2×2 = 24 Marks)

Q.3.(i)(a) Solve a problem to create a C++ program that calculates the area of a rectangle using classes and objects.

OR

(**b) Draw** the class hierarchy for a university system including classes like *Person*, *Student*, *Professor*, and show inheritance.

(ii)(a) Modify the following C++ class to include a destructor that displays a message when an object is destroyed:

```
cpp
CopyEdit
class Test {
public:
    Test() {
        cout << "Constructor called";
    }
};
```

OR

(b) Compute the output of the given C++ program that uses function overloading and explain how the correct function is selected.

Q.4.(i)(a) Differentiate between compile-time polymorphism and run-time polymorphism in C++ with suitable examples.

OR

(**b) Outline** the process of multiple inheritance in C++ with an example program structure.

(ii)(a) Structure a C++ program using classes to manage a simple bank account system, highlighting encapsulation and data hiding.

OR

(**b) Find coherence** between abstraction and encapsulation concepts in C++ programming, explaining how they complement each other.

SECTION – C

NOTE: Attempt all questions. (12×2 = 24 Marks)

Q.5.(a) Plan and **design** a C++ program that implements a library management system using classes and inheritance. Your program should allow adding new books, issuing books, and returning books.

OR

(b) Generate a model in C++ that simulates an online shopping cart system. Use appropriate classes for *Products*, *Cart*, and *User*, and demonstrate how object-oriented concepts like *encapsulation*, *inheritance*, and *polymorphism* are applied.

Q. 6.(a) Generate an application in C++ to **simulate** the working of a **Parking Lot Management System**. Design classes for vehicles, parking spaces, and parking lots, and show how the **abstract class** concept can be applied.

OR

(b) Produce a **Simple Inventory System** that uses **multiple classes** such as Product, Inventory, and Sales to keep track of items in stock and their prices. Implement **operator overloading** to update product quantities and prices.